








Advanced Time Series Analysis and Forecasting

Seminar 15: Review and project defence



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Acronyms used in this material

- **ACI**: Adaptive Conformal Inference
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CW**: Clark–West (test for nested models)
- **DM**: Diebold–Mariano (test)
- **EODHD**: EOD Historical Data (market-data provider)
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **QML**: Quasi-Maximum Likelihood
- **RMSE**: Root Mean Squared Error
- **RMSPE**: Root Mean Squared Prediction Error
- **SVAR**: Structural Vector AutoRegression
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and seminar structure

- Question of the seminar
  - **can you use the tools of the whole course quickly and defend a replication result under questioning?**
- The seminar is a project clinic on Chapters 0–14
- Outputs to produce
  - the answers to the A exercises
  - for each B exercise, the answers you would give at the defence, with the numbers that support them
  - the pre-registration of your project (C1) and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the long-run variance of a mean (Ch. 0); a DM test by hand (Ch. 1)	Solved, Proposed	A1
A3, A4	Cholesky responses (Ch. 3); the half-life of an equilibrium error (Ch. 4)	Solved, Proposed	A3
A5, A6	one Kalman step (Ch. 6); a two-state Markov chain (Ch. 7)	Solved, Proposed	A5
A7, A8	the Kupiec test (Ch. 9); GARCH persistence (Ch. 8)	Solved, Proposed	A7
A9, A10	the split-conformal quantile (Ch. 13); placebo p-values and Holm (Ch. 14, 0)	Solved, Proposed	A9
B1, B2	mock defences: Meese–Rogoff for EUR/RON; the term spread	Solved, Proposed	B1
B3, B4	mock defences: a leakage audit; a rule search on the BET	Solved, Proposed	B3
C1, C2	your pre-registration with a power calculation; what is wrong in an AI answer?	Proposed	B1, A1–A10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: short problems on paper, as at the defence; Part B: mock defences on replication outputs, in the notebook; Part C: your pre-registration and the critique of an AI answer

Data used

Series	Source	Frequency	Period
EUR/RON reference rate	BNR	monthly (end of month)	2005–2026
US real GDP, 10-year and 3-month Treasury yields	FRED	quarterly	1962–2025
BET index	EODHD	daily	2001–2012
Simulated data (leakage audit)	–	–	–

- In the notebook: `dm_test`, `lrv`, `ols_se`, `leakage_r2`, `stationary_bootstrap_index`, `p_required`; no account or key is needed

Prerequisites for Today (1/20): the long-run variance

- The **long-run variance** sets the precision of a mean of dependent data: $\mathrm{Var}(\bar{x}) \approx \Omega/T$

$$\Omega = \gamma_0 + 2\sum_{j\geq 1}\gamma_j$$

  - γ_j: autocovariance at lag j (γ_0: the variance); T: sample size
- **AR(1)** $x_t = \phi x_{t-1} + \varepsilon_t$, innovation variance σ^2

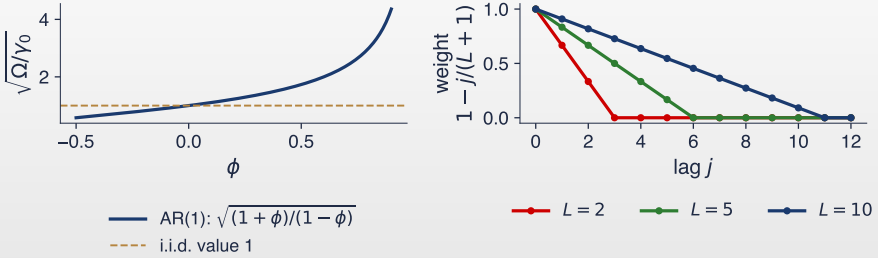
$$\gamma_0 = \frac{\sigma^2}{1-\phi^2}, \qquad \Omega = \frac{\sigma^2}{(1-\phi)^2}, \qquad \frac{\Omega}{\gamma_0} = \frac{1+\phi}{1-\phi}$$

  - $\sqrt{\Omega/\gamma_0}$: the factor by which i.i.d. standard errors are too small
- **Newey–West** estimator Newey and West (1987): weighted sum of sample autocovariances

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{L}\Big(1 - \frac{j}{L+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{T}\sum_{t>j}(x_t - \bar{x})(x_{t-j} - \bar{x})$$

  - L: number of lags (rule $L = \lfloor 4(T/100)^{2/9} \rfloor$, $\lfloor\cdot\rfloor$: integer part); $1 - j/(L+1)$: Bartlett weights

Prerequisites for Today (2/20): dependence and the precision of a mean



- Left: the factor $\sqrt{\Omega/\gamma_0}$ by which i.i.d. standard errors must be multiplied for an AR(1)
  - 1.36 for $\phi = 0.3$, 4.36 for $\phi = 0.9$; below 1 for $\phi < 0$ (negative autocorrelation)
- Right: the Bartlett weights $1 - j/(L+1)$ of $\hat{\gamma}_j$ in $\hat{\Omega}$; they decline linearly and vanish after lag L
  - the lag rule of the previous slide gives $L = 4$ for $T = 250$ and $L = 6$ for $T = 1000$; the weights keep $\hat{\Omega} \geq 0$

Prerequisites for Today (3/20): comparing two forecasts

- **DM test** Diebold and Mariano (1995): is the expected loss of forecast 1 equal to that of forecast 2?

$$d_t = L(e_{1t}) - L(e_{2t}), \qquad \mathrm{DM} = \bar{d} / \sqrt{\hat{\Omega}_d / P}$$

  - e_{it}: forecast error of model i; L: loss (e.g. squared error); $\bar{d}$: mean of d_t; $\hat{\Omega}_d$: its long-run variance; P: number of forecasts

- **HLN** correction Harvey, Leybourne and Newbold (1997) for small samples and horizon h

$$\mathrm{DM}^* = \mathrm{DM} \sqrt{\left(P + 1 - 2h + h(h-1)/P\right)/P}$$

  - compared with Student-t with $P - 1$ degrees of freedom (t_{P-1})

- **Clark–West** Clark and West (2007) for nested models (model 1 is the smaller one)

$$f_t = e_{1t}^2 - \left[e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2\right]$$

  - $\hat{y}_{it}$: forecast of model i; the bracket removes the noise of estimating the extra coefficients; one-sided test of $\mathbb{E} f_t \le 0$

Prerequisites for Today (4/20): accuracy measures and nested models

- **RMSE** of P forecasts and the ratio to a benchmark

$$\mathrm{RMSE} = \sqrt{\frac{1}{P}\sum_{t=1}^{P} e_t^2}, \qquad \text{ratio} = \frac{\mathrm{RMSE}_{\text{model}}}{\mathrm{RMSE}_{\text{benchmark}}}$$

  - ratio < 1: the model is more accurate; random walk for a log change: $\hat{y}_{t+1} = 0$
- **Clark–West** Clark and West (2007) statistic: the mean of the adjusted differential f_t over its standard error

$$\mathrm{CW} = \frac{\bar{f}}{\sqrt{\hat{\Omega}_f / P}}, \qquad \bar{f} = \frac{1}{P}\sum_{t=1}^{P} f_t$$

  - $\hat{\Omega}_f$: the long-run variance of f_t; reject the smaller model if $\mathrm{CW} > 1.645$ (one-sided, 5%)
- **Out-of-sample** R^2: positive when the forecast beats the training mean $\bar{y}_{\text{train}}$

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y}_{\text{train}})^2}$$

- t**-statistic** of a slope: $t = \hat{\beta}/\mathrm{se}(\hat{\beta})$, with a classical or a HAC standard error (Newey–West Newey and West (1987), L lags)

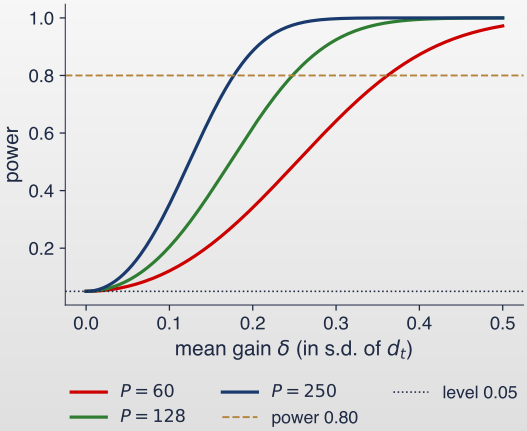
Prerequisites for Today (5/20): the power of a DM test

- Number of forecasts P needed to detect a mean gain δ with power π

$$P \approx \Omega_d \frac{(z_{1-a/2} + z_\pi)^2}{\delta^2}$$

  - δ: the mean gain, in standard deviations of d_t; a: the test level (e.g. 5%); z_q: the q-quantile of the standard Normal ($z_{0.975} = 1.96$, $z_{0.8} = 0.84$)
  - Ω_d: long-run variance of the standardised d_t; for an AR(1) differential with coefficient ρ, $\Omega_d = (1+\rho)/(1-\rho)$
- Reading
  - halving δ quadruples P; positive dependence ($\rho > 0$) multiplies P by Ω_d
  - a non-rejection with P far below this value says little about the absence of a gain

Prerequisites for Today (6/20): power curves of the DM test



- Power of a two-sided 5% test, i.i.d. differential ($\Omega_d = 1$)

$$\pi(\delta) = \Phi\big(\delta\sqrt{P} - 1.96\big) + \Phi\big(-\delta\sqrt{P} - 1.96\big)$$

  - Φ: the standard Normal distribution function; δ: the mean gain in standard deviations of d_t
- For $\delta = 0.2$: power 0.34 with $P = 60$, 0.62 with $P = 128$, 0.89 with $P = 250$
  - at $\delta = 0$ every curve equals the level 0.05
- With dependent d_t, replace $\sqrt{P}$ by $\sqrt{P/\Omega_d}$: the curves move to the right

Prerequisites for Today (7/20): structural VAR and cointegration

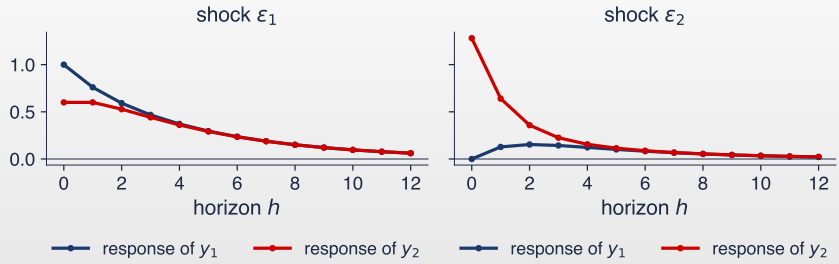
- **Recursive SVAR**: the reduced-form errors are a lower-triangular combination of structural shocks

$$u_t = B_0 \varepsilon_t, \qquad B_0 = \mathrm{chol}(\Sigma_u), \qquad \Theta_h = A^h B_0 \text{ (VAR(1))}$$

  - Σ_u: covariance of u_t; chol: Cholesky factor; A: VAR(1) coefficient matrix; Θ_h: responses at horizon h
  - 2×2: $b_{11} = \sqrt{\sigma_{11}}$, $b_{21} = \sigma_{21}/b_{11}$, $b_{22} = \sqrt{\sigma_{22} - b_{21}^2}$, with σ_{ij} the elements of Σ_u

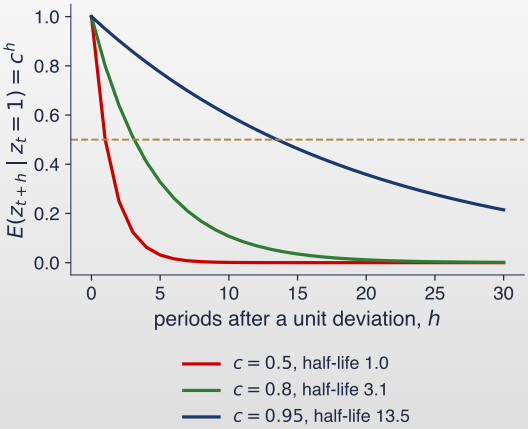
- **VECM** without short-run terms: $\Delta y_t = \alpha \beta' y_{t-1} + \varepsilon_t$
  - β: cointegrating vector; α: adjustment speeds; $z_t = \beta' y_t$: the equilibrium error, which follows an AR(1)
  - $z_t = (1 + \beta'\alpha) z_{t-1} + \beta' \varepsilon_t$; the half-life of a deviation: Prerequisites (9/20)

Prerequisites for Today (8/20): Cholesky impulse responses



- Illustration: $A = \begin{pmatrix} 0.7 & 0.1 \\ 0.3 & 0.5 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 1 & 0.6 \\ 0.6 & 2 \end{pmatrix}$; responses $\Theta_h = A^h B_0$ for $h = 0, \ldots, 12$
  - B_0: $b_{11} = 1$, $b_{21} = 0.60$, $b_{22} = 1.28$; $b_{12} = 0$, so y_1 does not respond on impact to ε_2 (right, $h = 0$)
  - element (i, k) of Θ_h: the response of y_i after h periods to a one-standard-deviation shock ε_k; stable VAR: all responses die out

Prerequisites for Today (9/20): the half-life of a deviation



- An AR(1) deviation $z_t = cz_{t-1} + \varepsilon_t$, $0 < c < 1$, decays on average as c^h

$$c^{h^*} = 0.5 \iff h^* = \frac{\ln 0.5}{\ln c}$$

  - h^*: the half-life, in periods; here 1.0, 3.1 and 13.5 for $c = 0.5, 0.8, 0.95$
- The same formula serves the equilibrium error of a VECM ($c = 1 + \beta'\alpha$) and a GARCH volatility shock ($c = \alpha + \beta$)
  - c close to 1: very slow return to equilibrium

Prerequisites for Today (10/20): the Kalman filter for a local level

- **Local level**: the observation is a random-walk level plus noise

$$y_t = \mu_t + \varepsilon_t, \qquad \mu_{t+1} = \mu_t + \eta_t$$

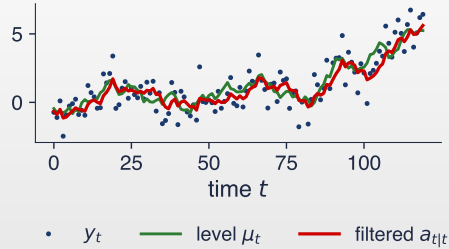
  - σ_ε^2, σ_η^2: variances of the observation and level noises; a, P: predicted level and its variance
- **One step**: prediction error variance, gain, update, next prediction

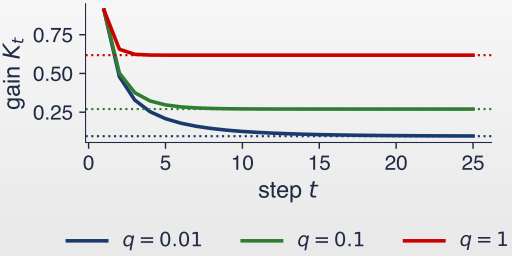
$$F = P + \sigma_\varepsilon^2, \quad K = P/F, \quad a_{t|t} = a + K(y - a), \quad P_{t|t} = P(1 - K), \quad P_{t+1} = P_{t|t} + \sigma_\eta^2$$

  - $K \in (0, 1)$: the share of the surprise $y - a$ added to the level
- **Steady state**: depends only on the signal-to-noise ratio $q = \sigma_\eta^2/\sigma_\varepsilon^2$

$$\bar{P} = \sigma_\varepsilon^2 \, \frac{q + \sqrt{q^2 + 4q}}{2}$$

Prerequisites for Today (11/20): the Kalman filter: a numerical example





- Left: a simulated local level with $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 0.1$; the filtered level follows the true level and smooths the noise
- Right: the gain K_t from a vague start ($P_1 = 10$) converges in a few steps to $\bar{K} = \bar{P}/(\bar{P} + \sigma_\varepsilon^2)$ (dotted)
  - $\bar{K} = 0.095, 0.270, 0.618$ for $q = 0.01, 0.1, 1$: a noisier level (larger q) gives more weight to the new observation
  - with K constant the filter is exponential smoothing: $a_{t|t} = (1 - \bar{K})a_{t-1|t-1} + \bar{K}y_t$

Prerequisites for Today (12/20): Markov chains and GARCH

- **Two-state Markov chain**: p_{ii} is the probability of staying in regime i from one period to the next

$$\mathbb{E}D_i = \frac{1}{1-p_{ii}}, \qquad \pi_1 = \frac{1-p_{22}}{2-p_{11}-p_{22}}, \qquad \lambda = p_{11}+p_{22}-1$$

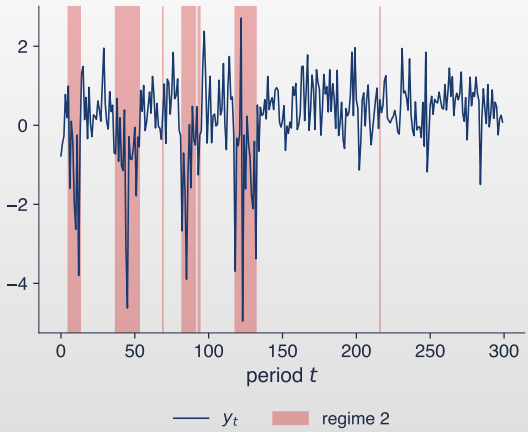
  - $\mathbb{E}D_i$: expected duration of regime i; π_1: long-run (ergodic) share of time in regime 1; λ: persistence of the chain

- **GARCH(1,1)**: today's variance from yesterday's squared shock and variance

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \qquad \sigma^2 = \frac{\omega}{1-\alpha-\beta}$$

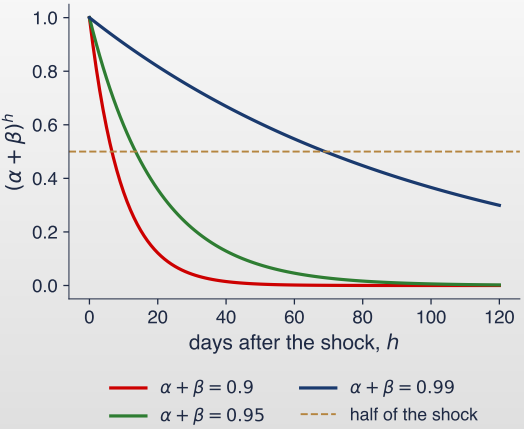
  - $\omega > 0$, $\alpha, \beta \geq 0$; ε_{t-1}: yesterday's shock; σ^2: unconditional variance (when $\alpha+\beta<1$)
  - persistence $\alpha+\beta$; the half-life of a volatility shock: Prerequisites (9/20) with $c = \alpha+\beta$

Prerequisites for Today (13/20): a simulated two-state Markov chain



- $p_{11} = 0.97$ (calm regime), $p_{22} = 0.90$ (turbulent regime, shaded); y_t has a lower mean and a larger variance in regime 2
- Expected durations 33.3 and 10.0 periods; ergodic share of regime 1 $\pi_1 = 0.769$
  - in this sample of 300 periods regime 1 occupies 0.817 of the time
  - $\lambda = 0.87$: the regime of today predicts the regime of the next periods
- The duration of a regime is geometric: $\Pr(D_i = d) = p_{ii}^{d-1}(1 - p_{ii})$, $d = 1, 2, \ldots$

Prerequisites for Today (14/20): the decay of a volatility shock



- GARCH(1,1) variance forecast h days ahead

$$\mathbb{E}_t\sigma^2_{t+h} - \sigma^2 = (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \sigma^2)$$

  - σ^2: the unconditional variance; a deviation of today's variance from σ^2 shrinks by the factor $\alpha+\beta$ each day
- Half-lives: 6.6, 13.5 and 69.0 days for $\alpha+\beta = 0.90$, 0.95, 0.99
  - near 1 the half-life explodes: small changes in the persistence change it a lot

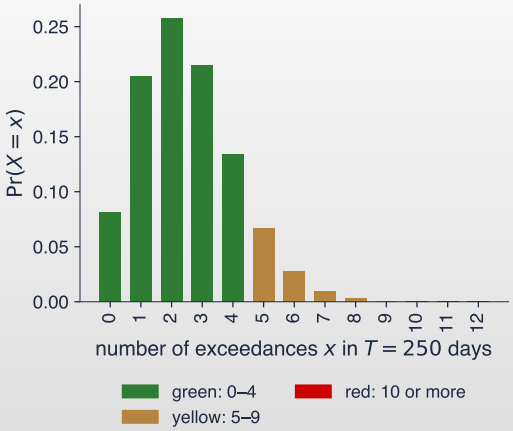
Prerequisites for Today (15/20): VaR backtests

- VaR 1%: $\mathrm{VaR}_\alpha = -q_\alpha$, with q_α the α-quantile of the return and $\alpha = 0.01$
- **Kupiec test**: x exceedances in T days; is the exceedance rate equal to α?

$$LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x}\alpha^x}{(1-\hat{\pi})^{T-x}\hat{\pi}^x} \sim \chi^2_1, \qquad \hat{\pi} = x/T$$

  - numerator: likelihood of the exceedances under the target rate α; denominator: under the observed rate $\hat{\pi}$
  - a large LR_{uc} (small p-value): too many or too few exceedances
- **Basel traffic light** for 250 days at 1%
  - green 0–4, yellow 5–9, red 10 or more exceedances

Prerequisites for Today (16/20): the binomial law of exceedances



- If the VaR is correct, the number of exceedances X in T days is Binomial(T, α)

$$\Pr(X = x) = \binom{T}{x} \alpha^x (1-\alpha)^{T-x}$$

  - $\binom{T}{x}$: the number of ways to place x exceedances among T days; $\mathbb{E}X = T\alpha = 2.5$ for $T = 250$, $\alpha = 0.01$
- The exact one-sided p-value of x_0 exceedances is the tail $\Pr(X \geq x_0) = \sum_{x=x_0}^{T} \Pr(X = x)$
- Basel zones for a correct model: green with probability 89.2%, yellow 10.8%, red 0.03%
  - a correct model falls in the yellow zone about one year in ten

Prerequisites for Today (17/20): conformal prediction

- **Split conformal** Romano, Patterson and Candès (2019): scores $S_1, \ldots, S_n$ on a calibration set, e.g. absolute residuals

$$\hat{q} = S_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil$$

  - $S_{(k)}$: the k-th smallest score; $\lceil \cdot \rceil$: rounding up; α: the allowed miss rate; the interval is $\hat{y} \pm \hat{q}$
- **Guarantee** under exchangeability (no ties)

$$1 - \alpha \leq \Pr(Y_{n+1} \in \hat{C}) \leq 1 - \alpha + \frac{1}{n+1}$$

  - $\hat{C}$: the prediction interval; exchangeable: the joint law does not change when the points are reordered
- **ACI** Gibbs and Candès (2021) for dependent data: the working level α_t is updated after each observation
  - $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$; γ: step; $\mathrm{err}_t = 1$ after a miss: a miss widens the next interval

Prerequisites for Today (18/20): searches, placebos and the defence

- **Multiple tests**: m tests at level a, p-values p_j, sorted $p_{(1)} \leq \cdots \leq p_{(m)}$
  - Bonferroni: reject when $p_j \leq a/m$
  - Holm: reject while $p_{(i)} \leq a/(m-i+1)$, stop at the first failure; both control the FWER (probability of at least one false rejection)
  - max-t (Reality Check logic White (2000)): the best t against the bootstrap law of the largest one (next slide)
- **Synthetic control placebo** Abadie, Diamond and Hainmueller (2015): rank of the treated unit among $J+1$ units
  - p-value $= \#\{j : r_j \geq r_1\}/(J+1)$; r_j: post/pre RMSPE ratio of unit j; the smallest p-value is $1/(J+1)$
- **Leakage**: any choice (selection, scaling, tuning) that uses data after the forecast origin Kaufman et al. (2012)
- At the defence: individual, three blocks of questions (code, results, chapters), no AI tools
  - a good answer: the number, its uncertainty, the assumption behind it, what would change it

Prerequisites for Today (19/20): a rule search in formulas

- **Moving-average rule** with window n: long the next day if the close is above its n-day average, otherwise cash

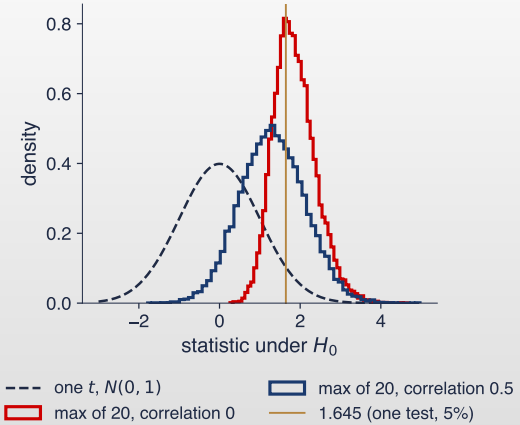
$$s_{n,t} = 1\Big\{P_t > \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}\Big\}, \qquad f_{n,t+1} = r_{t+1}\big(s_{n,t} - 1\big)$$

  - P_t: the close; r_{t+1}: the next log return; $f_{n,t+1}$: the return of the rule minus that of buy-and-hold
  - $t_n = \bar{f}_n / \mathrm{se}_{\mathrm{HAC}}(\bar{f}_n)$ for each of the K rules; the best rule has $t_{\max} = \max_n t_n$
- **Stationary bootstrap**: resample the days in blocks of random length, geometric with mean b, wrapping around the end
  - the same days for all K rules: the correlation of the rules is kept; blocks keep the dependence over time
- **Max-t p-value** with B resamples of the centred differentials $f_{n,t} - \bar{f}_n$

$$t^{*(b)}_{\max} = \max_n \frac{\bar{f}^{*(b)}_n - \bar{f}_n}{\mathrm{se}(\bar{f}_n)}, \qquad p = \frac{1}{B}\sum_{b=1}^{B} 1\big\{t^{*(b)}_{\max} \geq t_{\max}\big\}$$

  - the critical value is the 95% quantile of $t^*_{\max}$, not 1.645

Prerequisites for Today (20/20): the maximum of many t-statistics



- Simulated: 20 statistics $N(0, 1)$ under H_0, independent or with pairwise correlation 0.5
  - the maximum is far to the right of a single $N(0, 1)$
- 95% quantiles of the maximum: 2.79 (independent), 2.65 (correlation 0.5), against 1.645 for one test
  - with 20 independent tests at 5%, at least one rejection occurs with probability $1 - 0.95^{20} = 64\%$
  - the more correlated the rules, the closer the maximum stays to a single test

A1: the precision of a mean (Chapter 0) [Solved]

Task

- A series is AR(1) with $\phi = 0.6$ and $\sigma^2 = 1$. A team reports $t = 2.5$ for its mean with i.i.d. standard errors.
- 1. Compute γ_0 and Ω.
- 2. By what factor is the i.i.d. standard error too small?
- 3. Compute the corrected t and state the conclusion at 5%.
- Report: two numbers; one number; one number and one sentence.

Solution

- 1. $\gamma_0 = 1/(1 - 0.36) = 1.5625$; $\Omega = 1/(1 - 0.6)^2 = 6.25$
- 2. $\Omega/\gamma_0 = (1 + \phi)/(1 - \phi) = 4$: the standard error is too small by a factor of $\sqrt{4} = 2$
- 3. $t = 2.5/2 = 1.25 < 1.96$: not significant
- Interpretation: at the defence, the first question about any t-statistic is "which standard error?"

Back Exercise list

A2: a Diebold–Mariano test by hand (Chapter 1) [Proposed]

Solution discussed in class

Task

- Squared-error differentials d_t (benchmark minus model), $h = 1$: 0.8, −0.2, 0.5, 1.1, 0.3, −0.4, 0.9, 0.6, 0.2, 0.4. Model: A1.
- 1. Compute $\bar{d}$, $\hat{\gamma}_0$ and $\hat{\gamma}_1$ (divide by P).
- 2. Compute $\hat{\Omega}$ with the Bartlett kernel and $L = 1$, then DM and HLN.
- 3. Compare with the critical value of t_9 and state the conclusion.
- Report: three numbers; three numbers; one sentence.

Back Exercise list

A3: Cholesky responses (Chapter 3) [Solved]

Task

- A VAR(1) $y_t = Ay_{t-1} + u_t$ with $A = \begin{pmatrix} 0.5 & 0 \\ 0.2 & 0.4 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$, recursive ordering.
- 1. Compute B_0.
- 2. Compute $\Theta_1 = AB_0$ and $\Theta_2 = A^2 B_0$.
- 3. What changes if the ordering is reversed?
- Report: a matrix; two matrices; one sentence.

Solution

- 1. $b_{11} = \sqrt{4} = 2.0$, $b_{21} = 2/2 = 1.0$, $b_{22} = \sqrt{5-1} = 2.0$
- 2. $\Theta_1 = \begin{pmatrix} 1.00 & 0.00 \\ 0.80 & 0.80 \end{pmatrix}$, $\Theta_2 = \begin{pmatrix} 0.50 & 0.00 \\ 0.52 & 0.32 \end{pmatrix}$
- 3. Reversed, the first variable responds on impact to the second shock and the second no longer to the first; with $\sigma_{12} \neq 0$ all responses change: the ordering is the identifying assumption
- Interpretation: a defence question is always "why this ordering?"

Back Exercise list

A4: the half-life of an equilibrium error (Chapter 4) [Proposed]

Solution discussed in class

Task

- A bivariate VECM with $\beta = (1, -1)'$ and $\alpha = (-0.2, 0.05)'$, no short-run terms. Model: A3.
- 1. Derive the AR(1) of $z_t = \beta' y_t$ and its coefficient.
- 2. Compute the half-life of a deviation.
- 3. Which variable is weakly exogenous if $\alpha_2 = 0$, and how is that tested?
- Report: one equation and one number; one number; one sentence.

Back Exercise list

A5: one step of the Kalman filter (Chapter 6) [Solved]

Task

- Local level with $\sigma_\varepsilon^2 = 4$, $\sigma_\eta^2 = 1$; predicted state $a = 10$, $P = 2$; observation $y = 13$.
- 1. Compute v, F and K.
- 2. Compute $a_{t|t}$, $P_{t|t}$ and P_{t+1}.
- 3. Compute the steady-state $\bar{P}$ and gain.
- Report: three numbers; three numbers; two numbers.

Solution

- 1. $v = 3$, $F = 2 + 4 = 6$, $K = 2/6 = 0.333$
- 2. $a_{t|t} = 10 + K \cdot 3 = 11$; $P_{t|t} = 2(1 - K) = 1.333$; $P_{t+1} = 2.333$
- 3. $q = 0.25$: $\bar{P} = 2.562$, $\bar{K} = \bar{P}/(\bar{P} + 4) = 0.390$
- Interpretation: the filter moves one third of the way towards the surprise; the signal-to-noise ratio q alone sets the long-run gain

Back Exercise list

A6: a two-state Markov chain (Chapter 7) [Proposed]

Solution discussed in class

Task

- Regime 1 (expansion) with $p_{11} = 0.95$, regime 2 (recession) with $p_{22} = 0.80$, monthly data. Model: A5.
- 1. Compute the expected durations.
- 2. Compute the ergodic probability of regime 1 and the persistence λ.
- 3. Why can a likelihood-ratio test of one regime against two not use χ^2_2?
- Report: two numbers; two numbers; one sentence.

A7: the Kupiec test for VaR 1% (Chapter 9) [Solved]

Task

- A VaR 1% model has $x = 6$ exceedances in $T = 250$ days.
- 1. Compute the hit rate and the expected number of exceedances.
- 2. Compute LR_{uc} and its p-value; compare with the exact binomial tail $\Pr(X \geq 6)$.
- 3. Give the Basel zone and the conclusion.
- Report: two numbers; three numbers; one sentence.

Solution

- 1. 2.4%; expected 2.5
- 2. $LR_{uc} = 3.56$, $p = 0.059$; exact one-sided binomial $p = 0.041$
- 3. Yellow zone; the asymptotic test does not reject at 5%, the exact tail does: with $\alpha = 1\%$ and 250 days, report both
- Interpretation: backtests on a few tail days have little power; one exceedance changes the verdict

Back Exercise list

A8: GARCH persistence (Chapter 8) [Proposed]

Solution discussed in class

Task

- Daily returns in %, GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$. Model: A7.
- 1. Compute the persistence and the half-life of a volatility shock.
- 2. Compute the unconditional variance and the annualised volatility (250 days).
- 3. Why are Hessian standard errors of α misleading with fat-tailed returns?
- Report: two numbers; two numbers; one sentence.

Back Exercise list

A9: the split-conformal quantile (Chapter 13) [Solved]

Task

- Absolute residuals of 19 calibration points: 0.12, 0.85, 0.40, 1.30, 0.07, 0.66, 2.10, 0.95, 0.31, 1.75, 0.52, 0.22, 1.05, 0.48, 0.90, 1.42, 0.18, 0.73, 1.20; $\alpha = 0.1$.
- 1. Compute k and $\hat{q}$.
- 2. Give the coverage bounds of $\hat{y} \pm \hat{q}$.
- 3. Which assumption fails for daily returns, and what replaces it?
- Report: two numbers; two numbers; one sentence.

Solution

- 1. $k = \lceil 20 \cdot 0.9 \rceil = 18$: the 18th smallest score, $\hat{q} = 1.75$
- 2. Between 0.90 and $0.9 + 1/20 = 0.950$, marginally
- 3. Exchangeability fails with volatility clustering; ACI or weighted conformal restore long-run coverage, conditional coverage must be tested
- Interpretation: a guarantee is only as good as its assumption; say which one you rely on

Back Exercise list

A10: placebo p-values and Holm (Chapters 14 and 0) [Proposed]

Solution discussed in class

Task

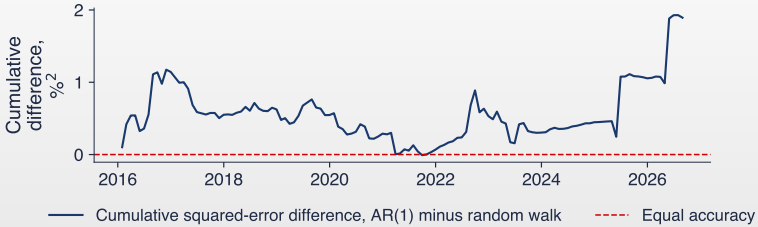
- A synthetic control with 19 donors; the treated unit has the largest post/pre RMSPE ratio. The team tests five outcomes with $p = 0.010, 0.020, 0.040, 0.200, 0.030$. Model: A9.
- 1. Compute the placebo p-value and the smallest attainable value.
- 2. How many outcomes are significant at 5% without correction, with Bonferroni and with Holm?
- 3. What should the pre-registration have said?
- Report: two numbers; three counts; one sentence.

B1: Meese and Rogoff for EUR/RON [Solved]

- **Question**: a team replicates Meese and Rogoff (1983): can an AR(1) beat the random walk for the monthly EUR/RON change? You are the examiner and the candidate at once
- **Inputs**: BNR reference rate, end of month, 100 $\times$ log change; rolling window of 60 months; 128 one-month forecasts from January 2016
- **Tasks**
  1. Produce the output: RMSE of both models, their ratio, DM–HLN and Clark–West.
  2. Defence question: why is Clark–West the right test here, and what does it test?
  3. Defence question: is the non-rejection evidence that the random walk is best? Use the power formula.
  4. Interpretation: write the one-sentence conclusion you would defend.
- **Report**: four numbers and two test statistics, two short answers, one sentence
- Notebook: section B1

Back Exercise list

B1: solution [Solved]



- RMSE: random walk 0.509, AR(1) 0.524; ratio 1.028; DM–HLN = 1.19 ($p = 0.235$); Clark–West = 1.11 (one-sided $p = 0.134$)
- Clark–West: the random walk is nested in the AR(1); under the null the estimated coefficients add noise to the larger model, which biases DM towards the random walk; CW corrects for it Clark and West (2007)
- Power: first-order autocorrelation of d_t -0.01; a gain of 0.2 standard deviations needs about 196 forecasts for 80% power, more than the 128 available
- Interpretation: "the Meese–Rogoff result holds for the leu over 2016–2026: the AR(1) does not beat the random walk, and our sample could only have detected a large gain"

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B2: the term spread and growth [Proposed]

- **Question**: a team replicates Estrella and Hardouvelis (1991) and reports "the spread predicts US growth, $t = 4.7$". Defend or correct it
- **Inputs**: FRED real GDP, 10-year and 3-month yields (quarterly averages), 1962–2025; $y_t = (400/4)\ln(\mathrm{GDP}_{t+4}/\mathrm{GDP}_t)$ on the spread; model: B1
- **Tasks**
  1. Estimate the slope with classical, Hansen–Hodrick-type (3 lags) and Newey–West standard errors.
  2. Defence question: why must the errors be at least MA(3)? Report the residual autocorrelation.
  3. Defence question: is the relation stable? Estimate it before 1989 and from 1989.
  4. Interpretation: rewrite the team's sentence so that it matches the evidence.
- **Report**: a slope, three standard errors and t-statistics, one autocorrelation, two subsample slopes, one sentence
- Notebook: section B2

Back Exercise list

B3: a leakage audit [Solved]

- **Question**: a team selects the 5 best of 100 candidate predictors by their correlation with the target over the whole sample, then reports a positive out-of-sample R^2 on the last 60 periods. Is the gain real?
- **Inputs**: a simulation with pure-noise predictors and target, $T = 240$, 1 000 replications; model: B1
- **Tasks**
  1. Run the team's protocol and the honest protocol (selection on the training sample only).
  2. Report the mean out-of-sample R^2 and the share of positive values for both.
  3. Defence question: which line of code leaks, and how is it fixed?
  4. Interpretation: what would you require before believing a reported R^2 of 10%?
- **Report**: four numbers, one line of code, one sentence
- Notebook: section B3

B3: solution [Solved]

- Team protocol: mean $R^2 = 5.9\%$, positive in 81% of replications (90th percentile 14.4%)
- Honest protocol: mean $R^2 = -13.3\%$, positive in 8%: the predictors are noise
- The leak: the correlations are computed on `rows = slice(0, T)`; the fix is `slice(0, n_train)`, and in a rolling design the selection is repeated at every origin
- Interpretation: a R^2 of 10% is within the range that the leak alone produces; ask for the selection code, a placebo with shuffled targets and a test period never touched

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B4: a rule search on the BET [Proposed]

- **Question**: a team reports "the 5-day moving-average rule beats buy-and-hold on the BET, $p = 0.006$". It tried 20 rules. Defend or correct it
- **Inputs**: BET daily log returns, 2001–2012 (2 993 days); rules "long if the close is above its n-day average", $n = 5, 10, \ldots, 100$; HAC t of the mean excess return; model: B3
- **Tasks**
  1. Compute the t-statistic of every rule and the naive p-value of the best.
  2. Compute the max-t p-value with the stationary bootstrap (mean block 10 days, 999 resamples).
  3. Defence question: why does the bootstrap resample the 20 series jointly, in blocks?
  4. Interpretation: what would you add before calling it a profitable rule?
- **Report**: 20 statistics, two p-values, one critical value, one sentence
- Notebook: section B4

C1: the pre-registration of your project [Proposed]

- **Question**: is your planned evaluation able to answer your question?
- **Inputs**: your team's landmark paper and data; the power formula of Prerequisites (5/20); model: B1
- **Tasks**
  1. Write the seven sections of the pre-registration in at most one page: question and hypotheses, paper to replicate, data, models, evaluation, robustness, team.
  2. Write the rule that decides "replicated", "partly" or "not replicated".
  3. Compute the evaluation length needed for 80% power at the gain reported by the paper, with ρ estimated on your training data.
  4. Interpretation: if the needed length exceeds your data, how do you change the question?
- **Report**: one page, one rule, one number, one sentence
- Notebook: section C1

Back Exercise list

C2: audit an AI answer [Proposed]

- A student asked an AI assistant how to prepare the replication project and its defence. The answer:
- `(a) Use the Diebold-Mariano test to compare your AR(1) with the random walk; it is valid for nested models.`
- `(b) With 19 donors, a placebo p-value of 0.01 is achievable if the effect is large.`
- `(c) If your result is not significant, the random walk is the true model.`
- `(d) For an AR(1) with phi = 0.6, i.i.d. standard errors of the mean are too small by a factor of 2.`
- `(e) Standardise the predictors once on the full sample before the walk-forward loop: it saves time and changes nothing.`
- `(f) Holm rejects more hypotheses than Bonferroni and controls the same family-wise error rate.`
- `(g) You may use the assistant during the oral defence if you declare it in AI_USE.md.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- Every number at the defence comes with its standard error, and the standard error with its assumption
- Nested models need Clark–West; a non-rejection needs a power calculation
- Overlapping targets and persistence enlarge standard errors; stability over subsamples is part of the evidence
- Selection on the whole sample creates out-of-sample gains from noise; a search needs a search-adjusted p-value
- Write the pre-registration and the verdict rule before the first estimate

After the seminar

- Lecture 15 develops each topic of today
  - the course map, one recap per chapter, the research workflow, the replication scoreboard
  - the failure modes, the oral defence and the AI policy
- Try the [Proposed] tasks in the notebook
- C1 is the first draft of your team's proposal and pre-registration
- Reading: Clemens (2017); Nosek et al. (2018); Diebold (2015)
- **The seminar is for practice and is not graded**

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